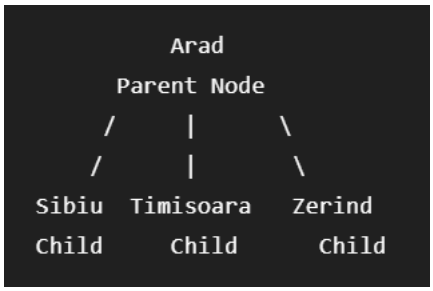
1. Define Parent Node with example.

A **parent node** is a node in a search tree that generates one or more new nodes by applying an action. The newly generated nodes are called **child nodes**.

Example: In the route-finding problem, **Arad** is the parent node of **Sibiu, Timisoara, and Zerind**.

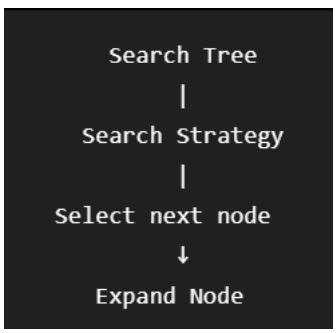


2. Define Search Strategy

A **search strategy** specifies the method used to decide **which node should be expanded next** from the frontier. Different search algorithms use different strategies for selecting the next node.

For example:

- **BFS** selects the shallowest node.
- **DFS** selects the deepest node.

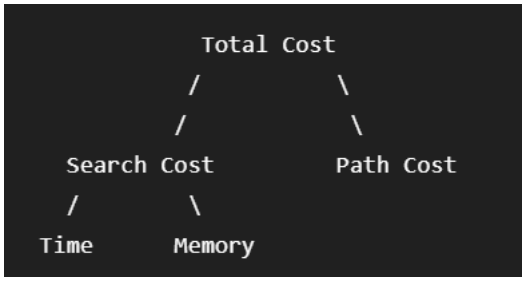


3. Describe cost in measuring problem solving performance

The performance of a search algorithm can be measured using **search cost** and **total cost**. Search cost mainly considers the time required and may also consider memory usage. Total cost combines the search cost and the path cost of the solution.

Formula:

Total Cost = Search Cost + Path Cost



4. Explain Zero-sum game with an example

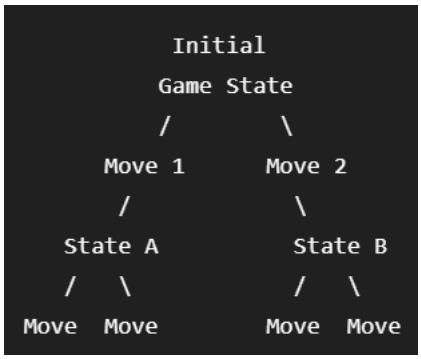
A **zero-sum game** is a game in which the gain of one player corresponds to the loss of another player, so the total payoff remains constant. **Chess** is an example of a zero-sum game.

For example, one player's winning outcome represents the other player's losing outcome.



5. Define Game tree

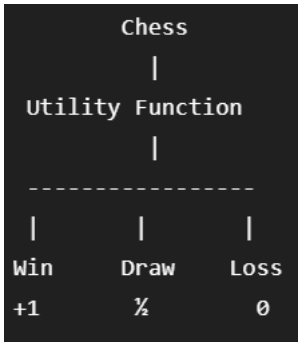
A **game tree** is a tree structure used to represent the possible states and moves in a game. The **nodes represent game states**, while the **edges represent possible moves**.



6. Explain Utility function to determine the Chess game

A **utility function** assigns a numerical value to the outcome of a game. It helps an AI agent determine whether a particular game outcome is desirable or undesirable.

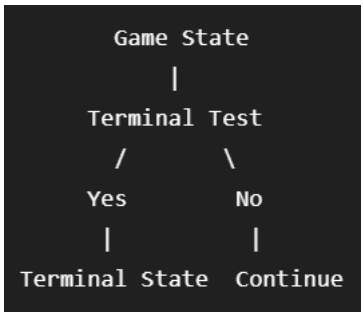
In chess, the possible outcomes include **win, loss, and draw**.



7. Which state is a Terminal State

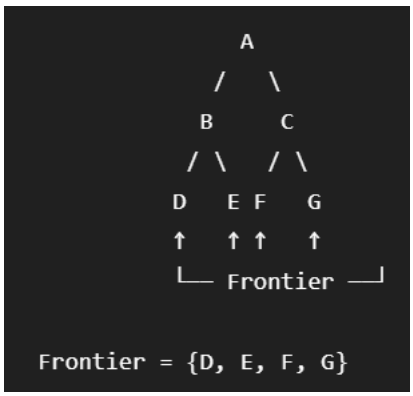
A **terminal state** is a state in which the game has reached its **end** and no further moves are possible. The terminal test is true for such a state.

Example: In chess, a completed game resulting in a win, loss, or draw is a terminal state.



8. Define Frontier

The **frontier** is the set of all **leaf nodes available for expansion** at a particular point in the search process. The search algorithm selects nodes from the frontier for further exploration.



9. Explain BFS and DFS in Uninformed search strategies

Breadth-First Search (BFS):

BFS expands the shallowest node first. It explores the search tree level by level and uses a FIFO (First-In, First-Out) queue.

Depth-First Search (DFS):

DFS expands the deepest node first. It follows one path as deeply as possible before backtracking and uses a LIFO (Last-In, First-Out) queue/stack.

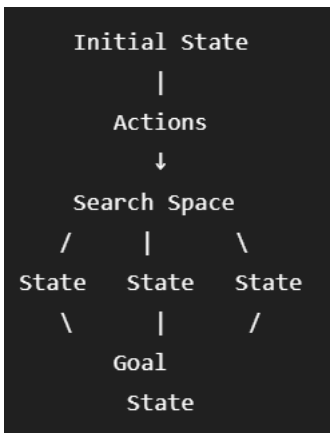


10. Describe Search algorithms

Search algorithms are methods used to find a solution by exploring different possible action sequences. The search starts from an **initial state** and continues by expanding nodes until a **goal state** is found or no more nodes are available.

Examples include:

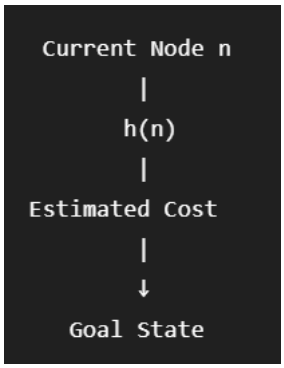
- Breadth-First Search
- Depth-First Search
- Uniform-Cost Search
- Greedy Best-First Search
- A* Search



11. Define heuristic function

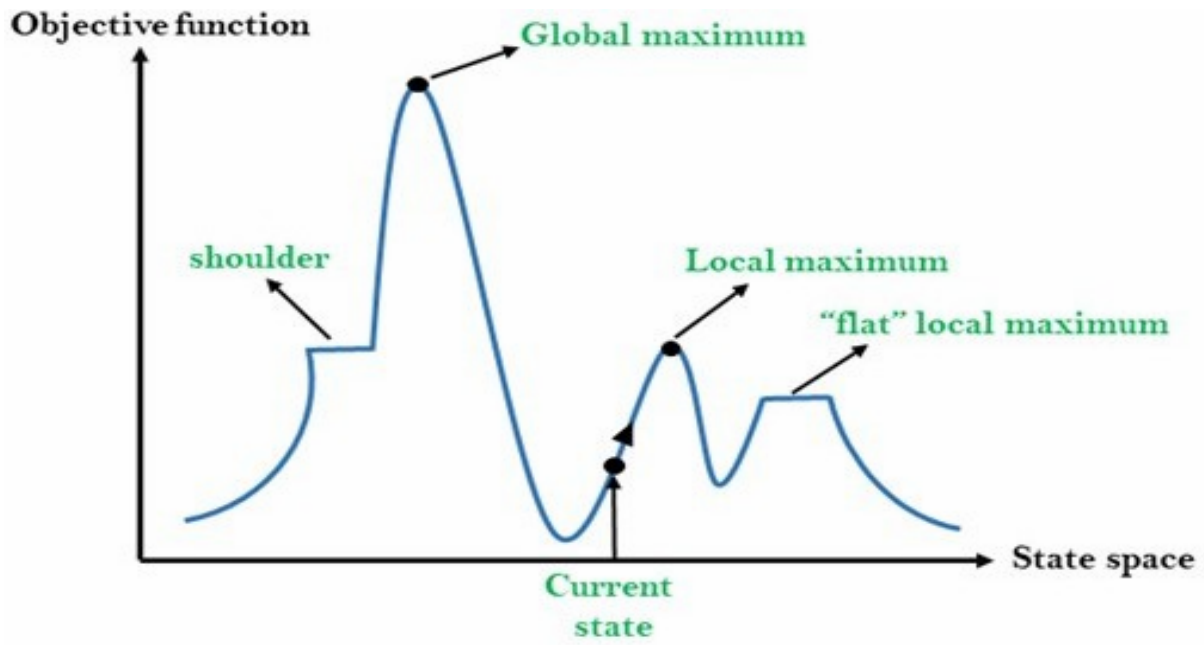
A **heuristic function** is represented by **h(n)**. It estimates the cost of the cheapest path from the current node **n** to a goal state. It is used in **informed search** to select a promising path.

Example: In route finding, straight-line distance to the destination can be used as a heuristic.



12. Define local maxima

A **local maximum** is a state that has a better value than all its neighboring states, but there is another state in the search space with an even higher value. It is one of the problems of the **hill-climbing algorithm**.



13. What is a search problem in AI?

A **search problem** in AI is a problem where an agent searches for a sequence of actions that transforms an **initial state** into a **goal state**. The possible action sequences form a search tree.

Example: Finding a route from one city to another.



14. Define uniformed search strategies.

Uninformed search strategies are search methods that do not use additional problem-specific knowledge or heuristic information. They use only the information available in the problem definition.

Examples:

1. Breadth-First Search (BFS)
2. Depth-First Search (DFS)

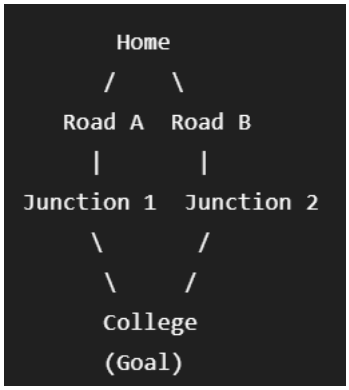
15. Give one example of a real-life problem that can be modelled as a search problem?

A **route-finding problem using Google Maps** is a real-life example of a search problem. The system searches through different possible roads to find a suitable route from the starting location to the destination.

Example: Finding a route from **Home to College**.

AI Search Model:

- **Initial State:** Home
- **Actions:** Take different roads
- **States:** Different locations/intersections
- **Goal State:** College
- **Path Cost:** Distance or travel time



16. Write one difference between BFS and DFS

BFS (Breadth-First Search)	**DFS (Depth-First Search)**
Explores nodes **level by level**.	Explores nodes **depth by depth**.
Uses a **FIFO Queue**.	Uses a **LIFO Stack**.
Generally requires **more memory** because it stores many nodes at each level.	Requires **less memory** because it mainly stores nodes along the current path.
Finds the **shallowest solution first** when a solution exists.	May find a **deeper solution before a shallower one**.